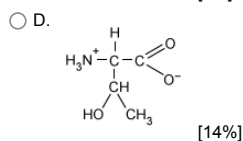
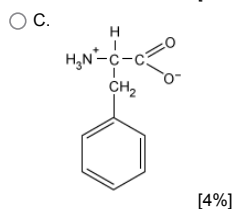
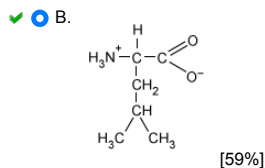
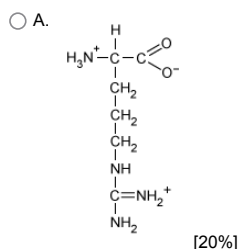
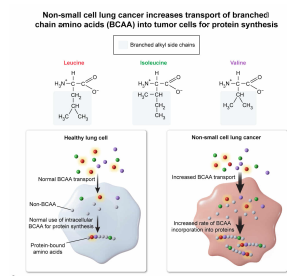


Given that non-small cell lung cancer (NSCLC) increases the uptake and use of plasma branched chain amino acids (BCAAs) for synthesis of tumor proteins, which structure below represents a BCAA that would be transported at a greater rate into NSCLC cells than into healthy lung cells?



Correct
60% answered correctly

Explanation:



The twenty **standard amino acids** found in proteins share **common structural elements**: amino, carboxyl, and R (side chain) groups attached to the alpha carbon of each amino acid. With the exception of **proline**, the backbone (ie, everything but the R group) of each amino acid is identical. It is variation in the R groups that give the amino acids their unique characteristics.

Three of the standard amino acids (leucine, isoleucine, valine) share the structural feature of having **branched alkyl side groups** and are **collectively known as the branched chain amino acids (BCAAs)**. Amino acids with branched chains that contain something other than alkyl groups are not counted among the BCAAs.

The question states that transformation of healthy lung cells into non-small cell lung cancer (NSCLC) cells causes the cells to increase transport of BCAAs into the cell for protein synthesis. The question asks for the amino acid that represents a BCAA that would be transported at a greater rate into NSCLC cells than into healthy lung cells. Although each of these amino acids possesses a branched side chain, only the **side chain of leucine** (shown in Choice B) is a **branched alkyl group**, so it is the only BCAA represented.

(Choice A) The **guanidinium group** of arginine is branched but it interrupts the chain of contiguous carbons, and the branches are formed by nitrogen atoms rather than the carbon atoms of an alkyl group. Therefore, arginine is not a BCAA.

(Choice C) The side chain of phenylalanine branches to form an **aromatic ring**. However, the bonds in an aromatic ring have partial double-bond character and are thus not the single bonds characteristic of an alkyl group.

(Choice D) Since threonine's hydroxyl group does not contain a carbon atom, it does not constitute a branched alkyl group.

Educational objective:

Branched chain amino acids have branched alkyl side chains.

Time spent: 0 sec
Subject:
Biochemistry

QID: 403332
System:
Amino Acids and Proteins